

1. "Merkle tree" is used in blockchain to:
  - a) Compress transaction data efficiently
  - b) Encrypt hash algorithms
  - c) Slow down transactions
  - d) Generate token supply

**Answer: a**

2. In a PoW system, mining difficulty adjusts based on:
  - a) Transaction fee
  - b) Network hash rate
  - c) Gas price
  - d) Block size

**Answer: b**

3. Byzantine Fault Tolerance ensures:
  - a) System continues to work despite faulty nodes
  - b) Blocks are reversible
  - c) All nodes act centrally
  - d) Miners are punished

**Answer: a**

4. Zero-Knowledge Proofs enable:
  - a) Privacy-preserving transaction validation
  - b) Node synchronization
  - c) Hash function creation
  - d) Network encryption only

**Answer: a**

5. Which blockchain protocol is known for scalability using fragments?
  - a) Polkadot
  - b) Ripple
  - c) Stellar
  - d) Dogecoin

**Answer: a**

6. In Ethereum 2.0, validators are chosen based on:
  - a) Computing power
  - b) Amount of stake
  - c) Random block selection
  - d) Number of nodes

**Answer: b**

7. The Lightning Network improves Bitcoin by:
  - a) Enabling off-chain transactions
  - b) Replacing miners
  - c) Adding new tokens
  - d) Reducing block size

**Answer: a**

8. Interoperability in blockchain allows:
- a) Different blockchains to communicate and share data
  - b) Data deletion
  - c) Token minting
  - d) Hash cloning

**Answer: a**

9. The “Trilemma” of blockchain refers to the difficulty in achieving:
- a) Security, scalability, and decentralization simultaneously
  - b) Fast mining
  - c) Token pricing
  - d) Node replication

**Answer: a**

10. Hash collisions in blockchain are:
- a) Extremely rare but theoretically possible
  - b) Common and easy to find
  - c) Always fixed automatically
  - d) Regular recalculations

**Answer: a**

11. Which of these ensures data integrity in blockchain?
- a) Hash chaining
  - b) Encryption
  - c) Decryption
  - d) Forking

**Answer: a**

12. What is the block time of Ethereum (PoS)?
- a) 1 minute
  - b) 12 seconds
  - c) 10 minutes
  - d) 1 second

**Answer: b**

13. What is the main advantage of sharding?
- a) Faster consensus
  - b) Scalability
  - c) Better encryption
  - d) Lower fees

**Answer: b**

14. Which cryptographic function is used for Ethereum hashing?
- a) SHA-256
  - b) Keccak-256
  - c) MD5

d) AES

**Answer: b**

15. In blockchain, what is "Sybil attack"?

- a) Creating fake nodes
- b) Deleting blocks
- c) Data corruption
- d) Network overload

**Answer: a**

16. What is zk-SNARK used for?

- a) Zero-knowledge proof
- b) Hash verification
- c) Mining
- d) Token generation

**Answer: a**

17. In Ethereum 2.0, what replaced miners?

- a) Validators
- b) Oracles
- c) Shards
- d) Nodes

**Answer: a**

18. What is the function of a blockchain "oracle"?

- a) It predicts the future price of cryptocurrencies
- b) It is a security system that audits smart contracts for bugs
- c) It is a service that provides external, real-world data to smart contracts
- d) It is the lead developer of a blockchain project

**Answer: c**

19. "Sharding" in blockchain is used to:

- a) Shrink block size
- b) Distribute data across multiple chains for scalability
- c) Encrypt data faster
- d) Increase mining difficulty

**Answer: b**

20. What problem do zk-Rollups solve in blockchain?

- a) They enable scalability by batching multiple transactions into one proof
- b) They slow down transaction time for better accuracy
- c) They handle mining distribution
- d) They replace nodes with oracles

**Answer: a**

**21. What is the main idea behind “state channels”?**

- a) Performing multiple transactions off-chain and settling the final state on-chain
- b) A layer for staking rewards
- c) A decentralized file storage method
- d) A network governance feature

**Answer: a**

**22. What is “slashing” in Proof of Stake networks?**

- a) A penalty for validators who act maliciously or stay offline
- b) A block compression technique
- c) A new type of encryption
- d) A token airdrop

**Answer: a**

**23. In a blockchain network, what is the role of a “light node”?**

- a) Stores only block headers to verify transactions efficiently without full data
- b) Stores every full block
- c) Validates smart contracts
- d) Issues new tokens

**Answer: a**

**24. What is MEV (Maximal Extractable Value)?**

- a) The maximum profit a validator can gain by reordering or including specific transactions in a block
- b) The total market value of a token
- c) The validator’s mining speed
- d) The reward for storing NFTs

**Answer: a**

**25. What is the purpose of “Cross-chain bridges”?**

- a) To allow tokens or data to be transferred between different blockchains
- b) To connect blockchain to traditional databases
- c) To duplicate nodes
- d) To create backup copies of blocks

**Answer: a**